Loan Default Analysis Report

SQL-Based Data Analysis Project

1. Introduction

What This Dataset Is About

This dataset contains detailed records of loan applications, borrower demographics, financial history, and repayment outcomes. Each row represents a unique loan account and includes attributes such as income, employment status, credit score, loan amount, loan purpose, and whether the borrower ultimately defaulted on the loan.

What Problem We Are Solving

The project focuses on analysing the loan portfolio data to identify key drivers of default risk and exposure. Using SQL for data querying and Power BI for visualisation, the dashboard highlights how borrower characteristics such as credit score, income group, and loan size influence default behaviour.

The goal of this analysis is to uncover high-risk segments and provide actionable insights for better risk management and lending decisions.

2. Objectives / Analysis Questions

The following core questions drive this analysis:

- Does income level affect the likelihood of loan default?
- Which borrower segments have the highest default rates?
- Is there a correlation between loan amount and default probability?
- How does credit score relate to loan default?
- What combination of factors drive the highest portfolio risk?

3. Dataset Information

Source

Kaggle – csv form

Total Records

~255K rows

Key Columns

- Loan Amount
- Income
- Credit Score
- Default

4. Tools used

Excel – Raw dataset handling
SQL – Data cleaning, aggregation and KPI calculation
Power BI – Dashboard creation & visualisation

5. Key KPIs

- Total Loans – 255K
- Default Rate – 11.61%
- High-Risk Exposure – 3.86%
- Average Loan Value – 128K

6. Dashboard Overview

The dashboard provides a structured view of portfolio risk:

- Top KPIs – Quick summary of portfolio health.
- Credit Risk Analysis – Default rates across credit groups.
- Income Distribution – Exposure across borrower's income segments.
- Risk Matrix – Combined impact of income and loan size.
- Heatmap Analysis – Interaction between income and credit score.
- Risk vs Exposure – Relationship between loan size and default behaviour.

7. Key Insights

- Low-credit borrowers have the highest default rate (12.8%), indicating strong correlation between credit quality and risk.
- High-income borrowers account for ~55.6% of total loan exposure.
- Highest risk observed in low-income borrowers with high loan amounts (~24% default rate).
- Overall default rate is 11-12% across all segments which is not healthy.
- Larger loan sizes drive both higher exposure (20.9bn) and higher default risk (14.5%).
- Combined effect of low income and low credit score significantly increases default probability (~18-24%).

8. Business Recommendations

- Tighten lending policies for low-income and low-credit borrowers.
- Introduce stricter approval checks for high loan amounts.
- Cap loan amount for high-risk borrowers.
- Focus on collection efforts for the high-risk segment.
- Monitor high-risk segments contributing disproportionately to defaults.
- Optimize portfolio by balancing the exposure across borrower categories.

9. Conclusion

This project demonstrates how combining data analysis with visualisation can uncover critical risk patterns in loan portfolios. The insights delivered can help financial institutions improve decision-making, reduce defaults, and manage portfolio exposure effectively.